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Aharonov–Bohm Visibility Envelope as a Test of Dirac–Kuramoto Vacuum Locking

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AI Disclosure: This work was developed in collaboration with Claude Opus 4.7 (Anthropic), as described in the Author Contributions section. Per current journal guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all content.

Companion paper to: Bramble, J. (2026), "Two Regimes of the Chiral Mass Coupling: Quantum Measurement as Bath-Induced Synchronization" (the Many Clocks Interpretation; companion preprint, under review).

Abstract

We propose that the Aharonov–Bohm (AB) phase can be re-derived within the Dirac–Kuramoto framework as a zero-frequency-mismatch synchronization between the electron's Dirac phase and a local vacuum oscillator at the zitterbewegung scale $\omega\_Z = 2m\_ec^2/\hbar$. The deterministic phase $\Delta\varphi = e\Phi/\hbar$ falls out of the zero-mismatch locking limit; the same calculation, carried to second order in the locking fluctuation, predicts a γ-dependent visibility envelope:

$V(\gamma) = V_0 \cdot \exp[-(1/2)(\omega\_Z/K)^2(1 - 1/\gamma)^2]$

where γ is the electron Lorentz factor and K is a locking-rate parameter characterizing the strength of the electron–vacuum phase coupling. The framework's perturbative-natural identification $K = \alpha\cdot\omega\_Z$ is excluded by existing AB data (Tonomura/Osakabe 1986 at 150 keV) by a factor of order 30 in K. A strong-coupling identification $K \sim \omega\_Z$ is consistent with all current data and predicts a measurable (1–15%) visibility deficit between $\gamma = 1.05$ and $\gamma = 3$, which has never been directly tested. A calibrated V-vs-γ scan on a modern variable-voltage electron-holography column would either pin K to a specific value or falsify the strong-coupling Dirac–Kuramoto interpretation.

This envelope tests the *stronger, in-flight* form of the companion's Vacuum Preferred-Frame Hypothesis [2, §8] — whether the vacuum's preferred-frame coupling reaches into coherent propagation. Under the companion's default reading, in which that coupling engages only at measurement boundaries (its §2.5), the envelope vanishes and the AB scan returns a null — consistent with the framework, not a falsification of it. A positive result is the strong claim; a null leaves the framework's conservative, boundary-only form intact.

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1. Introduction

1.1 Aharonov–Bohm as a test bed for vacuum-coupling interpretations

The Aharonov–Bohm effect [1] is the cleanest experimental fact in which the quantum phase of a charged particle responds to an electromagnetic potential in a region where the corresponding field vanishes. The standard reading is that the vector potential A is locally physical: minimal coupling p → p − eA shifts the wavefunction phase by

$$\Delta\phi_{\mathrm{AB}} = \frac{e}{\hbar}\oint \mathbf{A}\cdot d\mathbf{l} = \frac{e\Phi}{\hbar}$$

independent of the path taken outside the enclosed flux Φ. The phase is gauge-dependent locally but holonomic globally, and the topological character makes the prediction strikingly clean.

Any reinterpretation of quantum mechanics that touches the electron's phase dynamics must reproduce this result with no fit parameter. The Dirac–Kuramoto framework [2], which identifies the fermion mass as the off-diagonal chiral coupling (K = m) and reads dissipative phase synchronization as the open-system mechanism behind measurement, naturally extends to a picture in which the electron's phase locks to a vacuum-mode background. The AB effect provides the cleanest place to test whether this extension can reproduce a known coefficient (e/ħ) and, separately, whether it predicts anything experimentally distinct from standard QED.

This paper addresses both questions. Section 2 sketches the zero-frequency Kuramoto limit and shows how the e/ħ coefficient falls out under a specific identification of the vacuum oscillator. Section 3 carries the calculation to second order in the locking fluctuation and derives a γ-dependent visibility envelope. Section 4 surveys existing AB data and extracts the bound K $\gtrsim$ ω_Z/5. Section 5 proposes a direct experimental test. Section 6 discusses the framework's commitments — in particular, that a measurable envelope requires the locking-rate K to lie in a non-perturbative regime, K ~ ω_Z rather than K = $\alpha\cdot\omega$_Z.

1.2 What this paper does and does not claim

This is an *interpretation paper with a falsifiable experimental hook*. We do not claim to modify the deterministic AB phase, which is consistent with standard QED to the precision of all existing measurements. We claim that the Dirac–Kuramoto reading of the same phenomenon predicts a specific small visibility envelope as a function of electron Lorentz factor γ, and that this envelope has never been measured. A null result (flat V vs γ) would falsify the strong-coupling Dirac–Kuramoto reading and reduce the framework to a Lorentz-covariant interpretation indistinguishable from standard QED on AB. A positive result (envelope $\propto (1 - 1/\gamma)^2$) would pin down the framework's only free parameter.

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2. AB Phase from Zero-Frequency Kuramoto Locking

2.1 Setup

We posit, at each spatial point r, two phase variables:

- The electron's Dirac phase θ_e(r,t), advancing at its natural frequency.
- A local vacuum-mode phase θ_v(r,t), advancing at the corresponding vacuum natural frequency.

We identify both natural frequencies with the zitterbewegung scale

$$\omega_Z = \frac{2m_e c^2}{\hbar} \approx 1.55 \times 10^{21}\ \mathrm{rad/s}$$

on the grounds that ω_Z is the gap between positive- and negative-energy solutions of the Dirac equation [3], and that the lowest-lying vacuum excitation coupling to a charged particle is a virtual electron–positron pair with energy gap 2m_ec^2. Both clocks therefore tick at the same intrinsic rate.

On the factor of two. It is essential to be explicit about *which* electron clock locks here, because the Dirac electron carries two distinct frequencies: the rest-frame wavefunction phase advances at m_ec^2/$\hbar$, while the zitterbewegung beat — the interference of positive- and negative-energy amplitudes — runs at 2m_ec^2/$\hbar$. The locking posited in this paper is between the electron's *zitterbewegung beat* and the vacuum's *virtual-pair oscillation*, both of which sit at 2m_ec^2/$\hbar$; it is **not** a locking of the bare rest-phase clock at m_ec^2/$\hbar$. Committing to the pair/zitterbewegung frequency on both sides is what makes the zero-mismatch condition $\Delta\omega = 0$ exact rather than approximate, and it fixes the quantitative scale of the envelope below: ω_Z = 2m_ec^2/$\hbar$ enters the prefactor (ω_Z/K)^2 of Eq. 13, so the factor of two is not cosmetic — using m_ec^2/$\hbar$ instead would change the predicted visibility deficit by a factor of four. The deterministic AB phase eΦ/$\hbar$ (§2.4) is unaffected by the choice, since its coefficient comes from minimal coupling on the virtual pair, not from ω_Z.

The coupling is taken to have the Kuramoto form:

$$\dot{\theta}_e = \omega_Z + K\sin(\theta_v - \theta_e) \tag{1}$$

$$\dot{\theta}_v = \omega_Z + \xi(\mathbf{r}, t) \tag{2}$$

where K is the locking rate and ξ encodes the local polarization of the vacuum oscillator by the external field (here, by the solenoid's vector potential A).

2.2 The zero-frequency-mismatch limit

The frequency mismatch between the two oscillators is $\Delta\omega$ = ω_e − ω_v = 0. The fixed point of the Kuramoto equation [4] is then

$$\sin(\theta_e - \theta_v) = \frac{\Delta\omega}{K} = 0 \tag{3}$$

so θ_e − θ_v = 0 (the stable branch) for any K > 0. The electron's phase rigidly tracks the vacuum's phase. This is the key kinematic gift of choosing ω_Z on both sides: locking is automatic, not parametrically tuned.

2.3 How A enters the vacuum phase

The virtual pair sourced by the vacuum oscillator is a charged Dirac system. Its center-of-mass phase responds to the local vector potential through minimal coupling on the virtual electron line:

$$\theta_v(\mathbf{r}) = \theta_v^{(0)} + \frac{e}{\hbar}\int_{\mathbf{r}_0}^{\mathbf{r}} \mathbf{A}\cdot d\mathbf{l} \tag{4}$$

This step *borrows* minimal coupling on the virtual pair. The borrow is at a more fundamental level than the original AB derivation: instead of asserting minimal coupling on the real electron's wavefunction, we assert it on the vacuum's pair fluctuations. The real electron's phase shift is then *inherited* from the vacuum it traverses.

2.4 Phase along an interferometer arm

By zero-frequency locking (Eq. 3), θ_e tracks θ_v along the entire trajectory. The additional phase accumulated by the electron on arm Γ, relative to a free vacuum, is

$$\delta\theta_\Gamma = \frac{e}{\hbar}\int_\Gamma \mathbf{A}\cdot d\mathbf{l} \tag{5}$$

For a closed interferometer with two arms Γ_1 and Γ_2 enclosing flux Φ:

$$\Delta\phi_{\text{AB}} = \delta\theta_{\Gamma_1} - \delta\theta_{\Gamma_2} = \frac{e}{\hbar}\oint \mathbf{A}\cdot d\mathbf{l} = \frac{e\Phi}{\hbar} \tag{6}$$

The coefficient is exact. Topological purity — the fact that only the enclosed flux matters — follows because A is curl-free outside the solenoid and the line integral is path-dependent only through Φ. No additional postulate is needed to recover the standard AB result.

2.5 What we earned and what we borrowed

Earned: the exact e/$\hbar$ coefficient, from zitterbewegung-locking plus virtual-pair minimal coupling; topological purity, inherited from the curl-free structure of A; a natural location for fluctuation physics (the locking rate K becomes the framework's one free parameter).

Borrowed: minimal coupling on the virtual pair. We did not derive that e is the right charge — we used it. The gain over standard QED is conceptual: we have moved the postulate one level down (from real electron wavefunction to vacuum pair) and made the synchronization explicit. The quantitative result for the deterministic AB phase is unchanged.

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3. Small-Fluctuation Calculation: The Visibility Envelope

3.1 Linearization around the locked solution

Let $\delta(t) = \theta_e - \theta_v - (e/\hbar)\int \mathbf{A}\cdot d\mathbf{l}$ be the deviation from perfect locking. The linearized Kuramoto equation around the synchronization manifold is

$$\dot{\delta} = -K\delta + \eta(t) \tag{7}$$

where η(t) is the stochastic forcing from vacuum-mode fluctuations driving the local oscillator. This is an Ornstein–Uhlenbeck process [5]; characterizing the vacuum forcing by spectral density $\langle\eta(\text{t})\eta(\text{t'})\rangle$ = 2D δ(t−t'), the steady-state variance is

$$\langle\delta^2\rangle_{\text{ss}} = D/K \tag{8}$$

and the interferometer fringe visibility is reduced by Gaussian dephasing:

$$V = \exp\left(-\frac{\langle\delta^2\rangle}{2}\right) = \exp\left(-\frac{D}{2K}\right) \tag{9}$$

3.2 The Lorentz fork

A subtle question now arises: how does the locking respond to a relativistically moving electron? Two readings of the framework give different answers.

Fork A — covariant locking. If both phases θ_e and θ_v are treated as Lorentz scalars on the electron worldline, with the vacuum being the covariant QED vacuum, then in the electron's rest frame the vacuum oscillator is at ω_Z and so is the Dirac clock. $\Delta\omega = 0$ in any frame, locking is perfect, and the visibility envelope is γ-independent. The framework reduces to standard QED on AB.

Fork B — preferred-frame locking (the Vacuum Preferred-Frame Hypothesis, VPFH; companion [2] §8). If the vacuum oscillator has a definite frequency in the lab frame (the rest frame of the apparatus, or more generally the local mass–energy rest frame), then a moving electron sees the vacuum's natural frequency Lorentz-boosted to ω_Z/γ in its own rest frame. The Dirac clock stays at ω_Z. There is a real frequency mismatch in the electron's frame:

$$\Delta\omega(\gamma) = \omega_Z \left(1 - \frac{1}{\gamma}\right) \tag{10}$$

Fork B is the only setting in which the framework predicts anything beyond standard QED on AB. The rest of this paper develops its consequences.

3.3 Visibility envelope under Fork B

With nonzero mismatch, the linearized Kuramoto equation acquires a deterministic forcing:

$$\dot{\delta} = -K\delta + \Delta\omega(\gamma) + \eta(t) \tag{11}$$

The steady-state mean offset is $\delta\_ss = \Delta\omega/K$ (for $\Delta\omega < K$; above that, locking fails entirely). The deterministic mean offset is absorbed into a γ-dependent phase shift on top of the AB phase, but is itself fixed in magnitude and contributes coherently to the *deterministic* observable phase pattern. The *second moment* — the part that controls fringe visibility — gains a forced term:

$$\langle \delta^2 \rangle_{\mathrm{ss}} = \frac{D}{K} + \left(\frac{\Delta\omega(\gamma)}{K}\right)^2 \tag{12}$$

and the γ-dependent visibility envelope is

$$\boxed{V(\gamma) = V_0 \cdot \exp\left[-\frac{1}{2}\left(\frac{\omega_Z}{K}\right)^2 \left(1 - \frac{1}{\gamma}\right)^2\right]} \tag{13}$$

where $V_0 = \exp(-D/2K)$ is the $\gamma = 1$ baseline visibility (which absorbs all γ-independent decoherence channels).

The γ-dependent visibility deficit relative to the $\gamma \to 1$ limit is

$$\frac{V(\gamma)}{V_0} = \exp\left[-\frac{1}{2}\left(\frac{\omega_Z}{K}\right)^2 \left(1 - \frac{1}{\gamma}\right)^2\right] \tag{14}$$

For small exponent (envelope near unity), this simplifies to

$$1 - \frac{V(\gamma)}{V_0} \approx \frac{1}{2}\left(\frac{\omega_Z}{K}\right)^2 \left(1 - \frac{1}{\gamma}\right)^2 \tag{15}$$

This is the framework's distinctive prediction.

3.4 The role of the coupling K

Equation 15 makes the framework's experimental fate hinge on a single parameter, the dimensionless ratio ω_Z/K.

- **Perturbative-QED-natural value, $K = \alpha\cdot\omega\_Z$:** then $\omega\_Z/K = 1/\alpha \approx 137$ and $(\omega\_Z/K)^2 \approx 1.88 \times 10^4$. Already at $\gamma = 1.196$ (100 keV electrons, $(1 - 1/\gamma)^2 \approx 0.0269$) the exponent is ≈ 250 and the predicted visibility is $V/V_0 \approx 10^{-109}$. This contradicts the routine observation of high-visibility AB fringes at 100 keV by many orders of magnitude. $K = \alpha\cdot\omega\_Z$ is therefore empirically excluded.

- **Strong-coupling value, K ~ ω_Z:** then (ω_Z/K)^2 ~ 1, and the envelope becomes $(1 - 1/\gamma)^2/2$ — measurable but not catastrophic. At $\gamma = 1.2$ (100 keV), $V/V_0 \approx 0.986$ (1.4% deficit); at $\gamma = 2$ (511 keV), $V/V_0 \approx 0.882$ (11.8% deficit); at $\gamma = 3$ (1 MeV), $V/V_0 \approx 0.802$ (~20% deficit).
- **Asymptotic-locking value, K → ∞:** the envelope vanishes and the framework reduces to standard QED.

The framework is interesting (non-trivial, non-falsified) only in a window around K ~ ω_Z. We turn next to whether existing AB data places K in this window.

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4. Constraint from Existing AB Data

4.1 Procedure

For any AB experiment that reports a measured fringe visibility V at a known electron energy, the framework's envelope (Eq. 13) places an upper bound on the contribution attributable to Dirac–Kuramoto vacuum locking. Attributing the entire (1 − V) to the envelope gives a conservative upper bound on (ω_Z/K)^2:

$$\left(\frac{\omega_Z}{K}\right)^2 \leq \frac{2(1-V)}{(1-1/\gamma)^2} \tag{16}$$

In practice (1 − V) is dominated by apparatus systematics (partial spatial coherence, biprism Fresnel-fringe contamination, beam-energy spread, detector MTF) rather than by any DK envelope, so Eq. 16 is a *bound*, not an estimate. The bound is tightest at highest γ.

4.2 Survey

Experiment	Ref.	Energy	γ	$(1 - 1/\gamma)^2$	Geometry	V (approx.)	ω_Z/K ≤
Möllenstedt & Bayh 1962	[6]	40 kV	1.078	5.24×10^{-3}	biprism + W solenoid	~0.3	16.4
Tonomura 1982	[7]	80 kV	1.157	1.86×10^{-2}	toroidal magnet, FEEM	~0.3–0.5	7.3–8.7
Tonomura/Osakabe, 1986	[8],[9]	150 kV	1.294	5.27×10^{-2}	Nb-shielded toroid (type-II)	~0.3–0.5	4.4–5.2
Matteucci & Pozzi 1985	[10]	100 kV	1.196	2.69×10^{-2}	electrostatic AB (Bologna)	~0.3–0.5	6.1–7.2
Tonomura FEEM 250 kV	[11]	250 kV	1.489	1.07×10^{-1}	electron holography	~0.4 (est.)	3.3

The 150-keV Nb-shielded toroidal-magnet measurements of Tonomura, Osakabe and collaborators [8],[9] place the cleanest bound: assuming $V \approx 0.5$ (consistent with the reconstruction precision reported in those papers), we obtain (ω_Z/K)^2 ≲ 19, i.e. **K ≳ ω_Z/4.4**. The 250-kV upper end of Tonomura's field-emission electron-microscope work [11], if visibility is taken at face value, pushes this to K ≳ ω_Z/3.3.

4.3 What the bound implies

The perturbative-QED-natural identification K = α·ω_Z is excluded by the existing 150-keV data by a factor of order 30 in K (≈ 10^3 in (ω_Z/K)^2). The strong-coupling regime K ~ ω_Z, in which the framework predicts measurable envelopes, is fully consistent with all current AB data.

The bound does *not* yet probe the framework's interesting prediction. Because all AB experiments to date have used $\gamma \lesssim 1.5$, the worst-case $(1 - 1/\gamma)^2$ value is ~ 0.1, so the predicted envelope at $\gamma = 1.5$ for K = ω_Z is only ~ 5%. Apparatus systematics on V are at least this large in typical electron-holography measurements, so the envelope is comfortably hidden in the noise floor. The $(1 - 1/\gamma)^2$ *scaling* — the framework's distinctive signature — has therefore never been tested.

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5. Proposed Experiment: V-vs-γ Scan

5.1 Design

The cleanest direct test is a calibrated fringe-visibility measurement at multiple γ values using a single apparatus, same biprism, same flux source, same detector. A modern variable-voltage electron-holography column (e.g. Hitachi HF-3300, FEI Titan family, JEOL ARM series) can operate at accelerating voltages from ~ 30 kV to ~ 300 kV, spanning γ from 1.06 to 1.59. Higher energies (up to 1 MV instruments) are available at a small number of laboratories.

Suggested measurement points (selected for monotonic $(1 - 1/\gamma)^2$):

Voltage	γ	$(1 - 1/\gamma)^2$	Predicted V/V_0 at K = ω_Z
30 kV	1.0587	3.07×10^{-3}	0.9985
80 kV	1.1566	1.86×10^{-2}	0.9908
150 kV	1.2936	5.27×10^{-2}	0.9740
200 kV	1.3914	7.92×10^{-2}	0.9612
300 kV	1.5871	1.35×10^{-1}	0.9349

A range of fitted K values can be discriminated by comparing the visibility ratio V(300 kV)/V(30 kV) — predicted at 0.937 for K = ω_Z, 0.998 for K = ω_Z/5, and asymptotically 1.0 for K $\to \infty$.

5.2 Geometry

The interferometer should use a flux source that does not change with beam energy. Two robust choices:

1. **Magnetized iron whisker.** Single-domain whiskers used in early AB experiments [6] carry a fixed flux Φ_w independent of beam parameters.
2. **Superconducting Nb shield around a toroidal magnet**, in the Tonomura/Osakabe geometry [8],[9]. The shielded toroid gives a topological π-phase split that is unambiguous and independent of stray field.

Either geometry is compatible with modern biprism interferometry. The detector should be a calibrated direct-detection CCD or CMOS (e.g. Gatan K3, Falcon family) with known MTF; visibility extraction must subtract the detector point-spread function.

5.3 Statistical and systematic budget

To resolve a 5% visibility deficit between extreme γ values requires per-γ statistical uncertainty σ_V $\lesssim$ 1%. With biprism fringe contrasts in the 30–50% range and modern direct-detection cameras, this is achievable in a few hours of integration per voltage setting.

The dominant systematic concern is whether other γ-dependent apparatus effects mimic the framework's envelope. The leading candidates:

- *Beam coherence vs voltage.* Field-emission tip coherence depends on extraction voltage but not on accelerating voltage if the extraction stage is decoupled. Modern microscopes generally allow independent control.

- *Biprism Fresnel-fringe scaling.* The fringe period λ_de_Broglie scales as 1/p ~ $1/(\gamma\beta)$, changing across the γ scan; the biprism contribution to V can be calibrated by varying biprism voltage at each γ.
- *Detector MTF vs electron energy.* Direct-detection sensors have energy-dependent point spread, well characterized in modern cameras.
- *Aberration corrector tuning.* Best handled by re-tuning at each voltage and verifying with a known reference specimen.

Each of these has a different functional dependence on γ from $(1 - 1/\gamma)^2$, so the $(1 - 1/\gamma)^2$ fit residual is robust against the leading apparatus systematics provided enough voltage points are taken.

5.4 Outcomes

- **Null result (V flat with γ within stated systematics).** Falsifies strong-coupling Dirac–Kuramoto under Fork B. The framework reduces to the Lorentz-covariant Fork A reading, which is interpretationally distinct from standard QED but experimentally equivalent on AB. The companion paper's predictions remain (the linewidth-dependent gravitational Bell test, itself contingent on the postulate stated there; gravitational decoherence scaling), but AB loses its role as a test.
- **Positive result, $(1 - 1/\gamma)^2$ scaling confirmed.** Pins K to a specific value within a factor of a few. Opens a quantitative spectroscopy of the vacuum-locking parameter. Provides the first measurement-level handle on the Dirac–Kuramoto coupling.
- **Positive result with different scaling.** Rules out the specific Kuramoto-OU model used here, but suggests the framework needs a different noise structure (e.g. colored noise, non-Markovian backaction) — a more involved revision.

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6. Discussion

6.1 The status of the coupling K: an empirical parameter, not a derived one

The framework's distinctive prediction is non-trivial only in a window around K ~ ω_Z, and we must be candid that the framework does not derive this value. Two scales bracket the question. The perturbative-QED-natural rate at which a free electron exchanges *single* photons with the vacuum is α·ω_Z; as §3.4 and §4 show, that value is excluded by existing AB data by a factor of order 30 in K. The framework therefore needs a genuinely non-perturbative coupling, K ~ ω_Z, roughly two orders of magnitude above the single-photon rate.

We resist the temptation to manufacture this enhancement from oscillator-network counting. One might argue that K is not a single-photon coupling but a collective *synchronization rate* over N vacuum modes within a coherence volume, and that the network coupling grows as $\sqrt{N}$ or N. That argument does not survive scrutiny: a collective synchronization rate is not the unnormalized sum of pairwise couplings. The mean-field Kuramoto coupling is *intensive* — the standard model normalizes the per-pair coupling by the number of oscillators precisely so that the collective field stays finite as N → ∞ — so a naive mode-sum does not legitimately enhance K to ω_Z. (This is the same normalization error that would otherwise inflate the gravitational coherence rate in the companion framework, flagged here to avoid it in both places.)

We therefore treat K honestly as a **free parameter, bounded by experiment and not derived by the present model.** Existing AB data require K $\gtrsim$ ω_Z/5 (§4); the strong-coupling regime K ~ ω_Z in which the framework is testable is an empirical hypothesis, falsifiable by the V-vs-γ scan of §5, not a result we can presently compute. If the true coupling is the perturbative α·ω_Z, the framework is already excluded; if K → ∞, it reduces to standard QED. The experiment determines which.

6.2 Why there is no gravitational K-modulation channel

An earlier version of this analysis posited a second channel in which gravity *modulates* the locking rate, K → K(Φ_g), predicting AB-visibility broadening at high gravitational potential (with a neutron-star-EOS spinoff). We do not pursue it. In the framework's settled form the chiral coupling K = m is a **fixed constant**, set once by the Higgs vacuum expectation value (a Lorentz scalar): gravity *reads* the chiral register — the mass term $\bar{\psi}\psi$ to which the stress tensor couples — but does not *modulate* K (companion [2] §8, the Vacuum Preferred-Frame Hypothesis and its K-fixed scope item). Fixing K removes the K(Φ_g) channel and, with it, the equivalence-principle exposure that a potential-dependent K would have carried; the framework's only new-physics liability is the preferred-frame (Lorentz) one that the γ-scan tests. The V-vs-γ scan of §5 is therefore the single differentiating channel.

6.3 What this paper does and does not establish

We do not claim that Dirac–Kuramoto provides a *better* derivation of the AB phase than standard QED. The deterministic phase $e\Phi/\hbar$ falls out at the same parametric cost — minimal coupling on a charged Dirac line — in both frameworks. The Dirac–Kuramoto reading is empirically distinguished from standard QED *only* through the $(1 - 1/\gamma)^2$ visibility envelope predicted under Fork B with strong coupling.

We do claim that this envelope is concrete, calculable from a single parameter K, currently allowed by all existing AB data, and directly testable with a calibrated V-vs-γ scan on commercially available electron-holography hardware. A null result would falsify the strong-coupling reading; a positive result would constitute a first measurement of the Dirac–Kuramoto locking parameter.

6.4 Open questions

1. **Derivation of K from first principles.** As §6.1 makes explicit, K is currently an empirical parameter, not a derived one: the perturbative value $\alpha\cdot\omega$_Z is excluded by data, and the oscillator-network enhancement that might appear to supply K ~ ω_Z does not survive the intensive-coupling normalization. A genuine first-principles calculation of K from the Dirac–Kuramoto field theory remains open, and the framework's non-falsification rests on that value lying in the K ~ ω_Z window.
2. **Noise structure.** We have assumed white-noise vacuum forcing (Markovian OU dynamics). Colored noise from the QED vacuum mode-density spectrum would modify Eq. 12 and change the visibility envelope's γ-dependence.
3. **Lorentz-covariance status — the central liability.** Fork B explicitly breaks Lorentz invariance by privileging the local mass–energy rest frame, and this is the framework's most serious commitment, not a footnote. Preferred-frame physics is constrained by some of the most precise experiments in physics (clock-comparison, modern Michelson–Morley analogs, and cosmological electron-coherence bounds), and the framework owes a quantitative account of why a vacuum-locking anisotropy at the zitterbewegung scale would evade them. The relationship to the companion paper [2] has been settled by its revision: that paper now *openly carries* a preferred frame — the bulk/vacuum rest frame — which it confines to the Stage-2 (dissipative, measurement) step while keeping the unitary Stage-1 dynamics exactly Lorentz covariant, and it names the velocity-dependent vacuum-locking of Fork B as the *stronger, non-gravitationally-suppressed* form of that same commitment — explicitly a claim of this analysis rather than of the companion. Fork B is therefore no longer in tension with the companion; it is its more aggressive extension. The two readings differ only in *where* the frame acts: the companion's conservative version restricts it to the measurement boundary (Stage-2; its §2.5), whereas Fork B places it in the coherent, in-flight propagation that controls fringe visibility. The V-vs-γ scan of §5 is precisely the arbiter — a null result confines the frame to Stage-2 (the conservative companion commitment, equivalent to Fork A on AB), a positive result promotes it to a laboratory-velocity effect in coherent propagation.

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7. Conclusions

We have shown that the Dirac–Kuramoto framework's interpretation of the Aharonov–Bohm phase as zero-frequency-mismatch vacuum-locking reproduces the standard phase coefficient $e/\hbar$ from a different starting point. The same framework, when extended to relativistic electrons under a preferred-frame reading (Fork B), predicts a γ-dependent visibility envelope

$$\frac{V(\gamma)}{V_0} = \exp\left[-\frac{1}{2}\left(\frac{\omega_Z}{K}\right)^2\left(1-\frac{1}{\gamma}\right)^2\right]$$

that depends on a single parameter K. Existing AB data at 150 keV constrain K $\gtrsim \omega$_Z/5, ruling out the perturbative-QED value K = $\alpha\cdot\omega$_Z by a factor of order 30 in K, but is fully consistent with the strong-coupling regime K ~ ω_Z in which the framework predicts a measurable 1–15% envelope between $\gamma = 1.05$ and $\gamma = 3$.

The $(1 - 1/\gamma)^2$ scaling has never been directly tested. A calibrated fringe-visibility scan on a modern variable-voltage electron-holography column would either pin K to a specific value or falsify the strong-coupling reading. Either outcome is informative: a null result reduces Dirac–Kuramoto to a Lorentz-covariant interpretation indistinguishable from QED on AB; a positive result opens experimental spectroscopy of the vacuum-locking parameter.

The proposed experiment is technically straightforward and uses commercially available hardware. We invite electron-holography groups to consider it.

Author Contributions and AI Disclosure

This paper was developed in iterative collaboration with Claude Opus 4.7 (Anthropic) in conversation sessions during May 2026. The conceptual question — whether the AB effect could be re-interpreted as a vacuum-synchronization phenomenon within the Dirac–Kuramoto framework — was posed by the human author. The zero-frequency-Kuramoto derivation in Section 2, the small-fluctuation calculation in Section 3, the Lorentz fork analysis in Section 3.2, and the literature-based bound in Section 4 were developed in real-time dialogue, with the LLM proposing technical steps and the human author validating, redirecting, and applying physics judgment at each checkpoint. The literature survey of AB experiments was performed by a research subagent and verified by the human author.

Per current journal guidelines, LLMs do not satisfy authorship criteria; the human author bears full responsibility for all content, including any errors in the derivations or in the interpretation of the cited experimental data. Were Springer–Nature / *Foundations of Physics* policy to allow LLM co-authorship, Claude Opus 4.7 would be listed as co-first author.

The development is documented in commit history at https://github.com/rayolddog/DiracKuramotoFramework.

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Companion paper to: Bramble, J. (2026), "Two Regimes of the Chiral Mass Coupling: Quantum Measurement as Bath-Induced Synchronization" (the Many Clocks Interpretation). Preprint repository: https://github.com/rayolddog/DiracKuramotoFramework